

WHEELER-SACK AAF, NY, USA

WMO: 743700

Lat: 44.050N

Lon: 75.733W

Elev: 688

StdP: 14.33

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14715

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.6	-4.9	-15.6	2.4	-9.0	-10.6	3.2	-3.2	29.8	39.3	26.6	33.3	6.9	140	0.589

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	86.3	71.0	83.8	69.7	81.3	68.3	74.7	81.1	72.9	79.7	71.3	77.9	10.5	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.9	125.3	77.3	71.1	117.7	76.1	69.3	110.5	75.2	38.7	81.4	37.0	80.3	35.6	78.2	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	24.8	20.8	18.6	DB	-19.3	90.5	6.1	2.7	-23.7	92.4	-27.2	94.0	-30.6	95.5	-35.0	97.4
(5)				WB	-19.5	77.9	6.0	2.6	-23.8	79.8	-27.4	81.3	-30.7	82.7	-35.1	84.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.6	18.5	21.1	30.1	43.4	56.2	65.0	69.5	68.0	61.0	49.1	37.0	26.1
	DBStd	20.26	13.99	11.77	11.35	9.43	8.37	6.73	5.40	5.48	7.72	8.84	10.03	11.20
	HDD50	3966	978	808	624	242	33	0	0	8	128	404	741	
	HDD65	7542	1442	1228	1083	651	295	82	18	33	167	498	840	1205
	CDD50	2344	1	0	7	44	225	452	606	559	336	99	14	1
	CDD65	444	0	0	0	3	22	83	159	127	46	4	0	0
	CDH74	3437	0	0	4	34	229	676	1231	885	354	24	0	0
	CDH80	793	0	0	0	5	44	163	305	194	81	1	0	0
Wind	WSAvg	9.1	10.2	10.2	10.4	10.6	9.4	8.1	7.8	7.3	7.8	8.8	9.5	9.5
Precipitation	PrecAvg	38.7	2.9	2.4	2.6	3.1	3.3	3.2	3.6	3.1	3.6	4.1	3.6	3.3
	PrecMax	48.3	6.1	4.2	4.8	5.7	5.7	6.9	7.8	4.6	6.8	7.2	5.7	5.6
	PrecMin	30.8	1.0	1.0	1.1	0.4	1.3	0.7	1.3	0.9	1.9	0.7	1.7	1.5
	PrecStd	4.2	1.3	0.7	1.0	1.4	1.3	1.5	1.5	1.2	1.2	1.8	1.0	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.0	54.3	68.2	78.6	85.0	88.4	89.8	88.5	87.1	77.3	66.1	57.2
		MCWB	53.0	47.7	58.3	61.4	68.8	71.7	74.6	71.5	70.5	64.0	57.3	52.3
	2%	DB	46.4	46.0	57.4	71.2	79.9	84.3	86.3	84.4	81.9	71.8	61.1	49.8
		MCWB	43.9	41.5	48.7	55.4	64.3	69.4	71.9	70.1	69.1	62.5	53.6	46.9
	5%	DB	41.0	41.2	51.8	64.2	75.9	81.4	83.6	81.8	78.2	67.2	56.7	45.2
		MCWB	38.5	37.6	45.5	52.3	62.0	68.0	70.2	68.7	67.4	59.4	51.2	42.6
	10%	DB	36.6	36.9	45.8	59.3	71.7	77.8	80.9	79.2	74.3	63.1	52.1	40.6
		MCWB	34.4	34.3	40.8	49.7	59.7	65.9	68.5	68.0	65.2	56.9	47.9	38.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.2	48.3	59.6	64.2	71.7	75.6	77.8	76.8	74.6	69.0	59.2	53.6
		MCDB	55.6	52.1	69.3	72.5	81.2	83.5	84.7	81.6	80.9	71.5	63.1	56.6
	2%	WB	44.8	42.4	51.2	59.1	68.4	72.9	75.1	73.8	71.7	64.1	55.3	46.9
		MCDB	46.2	45.6	56.4	66.5	75.7	80.2	81.4	79.6	78.4	68.8	59.8	48.5
	5%	WB	38.7	38.0	45.8	55.1	65.6	70.8	73.1	72.1	69.1	60.4	51.7	42.7
		MCDB	40.7	41.0	50.5	62.8	72.6	77.6	79.8	78.0	75.2	65.5	55.9	45.2
	10%	WB	34.5	34.6	41.0	50.4	62.2	68.5	71.3	70.2	66.9	57.3	47.8	38.1
		MCDB	36.4	36.9	45.7	57.9	68.4	74.9	78.0	75.9	72.8	62.4	51.6	40.3
Mean Daily Temperature Range	5% DB	MDBR	16.3	16.4	16.2	18.8	19.9	18.7	18.3	18.8	19.6	16.6	14.6	14.0
		MCDBR	17.4	18.0	23.2	27.5	24.9	22.8	21.0	21.7	22.7	21.8	21.1	17.0
		MCWBR	16.2	15.4	16.2	15.8	12.9	11.1	9.6	10.1	11.5	13.3	15.4	14.8
	5% WB	MCDBR	17.2	17.8	21.6	24.1	21.0	18.7	17.4	17.2	19.2	19.2	19.2	16.7
		MCWBR	16.5	15.8	16.2	16.6	13.5	10.9	10.1	10.1	11.5	13.3	15.8	15.3
Clear-Sky Solar Irradiance	taub	0.289	0.294	0.325	0.354	0.380	0.412	0.425	0.414	0.364	0.346	0.308	0.283	
	taud	2.361	2.345	2.397	2.389	2.362	2.291	2.255	2.281	2.412	2.459	2.508	2.451	
	Ebn at Noon	265	285	290	289	283	273	267	266	273	263	259	258	
	Edh at Noon	25	30	33	36	38	41	42	40	32	27	22	21	
All-Sky Solar Radiation	RadAvg	422	695	1104	1391	1673	1788	1829	1619	1289	764	481	321	
	RadStd	46	74	93	170	174	147	161	79	114	76	55	43	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.79	N/A	N/A	N/A	N/A	N/A	-265	-378	N/A	N/A
(47)	Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page